

Name: Homework 1 Cover Page

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Problem set number: #1 Due date: 2/3/2026

Acknowledgement: "No help"

- 1) The dominant heat transfer mode for a spacecraft traveling outside the atmosphere is radiation. The spacecraft is in a vacuum outside the atmosphere so it is not in contact with a moving fluid or a solid preventing conduction and convection leaving radiation as the dominant mode. Once the spacecraft is traveling through the atmosphere the dominant heat transfer mode becomes convection because the spacecraft is traveling rapidly through a fluid ensuring the air operates like a forced cooling system.

2)

Known: $k_{\text{chip}} = 150 \text{ W/(m}\cdot\text{K)}$

$w = 5 \text{ mm}$

$t = 1 \text{ mm}$

$q_{\text{back}} = q_{\text{order}} = 0 \text{ W}$

$q_{\text{ss}} = 9 \text{ W}$

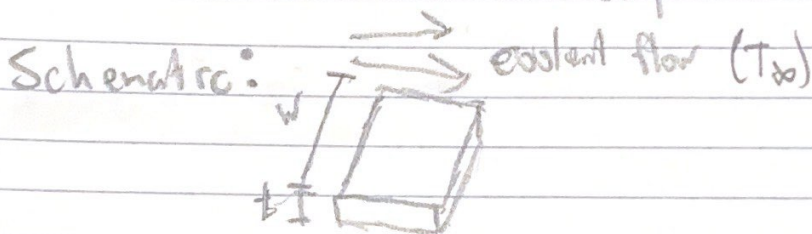
$q_{\text{rad}} = 0 \text{ W}$

$T_{\infty} = 20^{\circ}\text{C}$

Find: a) steady-state temperature difference between the front and back of the chip

b) Minimum heat transfer coefficient required to keep front of the chip $< 80^{\circ}\text{C}$

c) Discuss what convection processes can be used to cool the chip



Properties: All thermophysical properties used are listed in the known section

Assumptions: Steady state system

Analysis: Fourier's law;

Heat rate equation: $q_x = -kA \frac{dT}{dx}$

$$q_w = -150 \text{ W/(m}\cdot\text{K)} \cdot (0.005 \text{ m})^2 \cdot \frac{dT}{dx} \Rightarrow \frac{dT}{dx} = -2400 \text{ K/m}$$

$$2400 \text{ K/m} \cdot 0.001 \text{ m} = 2.4 \text{ K} =$$

$$\boxed{a) 2.4 \text{ K}}$$

Newton's Law of cooling:

$$\text{Heat rate: } q = h \cdot A \cdot (T_s - T_\infty)$$

$$9 \text{ W} = h \cdot (0.005 \text{ m})^2 \cdot (80^\circ \text{C} - 20^\circ \text{C})$$

$$h = 6000 \text{ W/(m}^2 \cdot \text{K)}$$

$$\boxed{b) 6000 \text{ W/(m}^2 \cdot \text{K)}}$$

c) Both liquid under forced convection and convection with phase change can provide a high enough heat transfer coefficient to cool the chip.

3)

Known: $\epsilon = 0.18$

$q_s = 0.76$

$G_s = 750 \text{ W/m}^2$

$T_\infty = 300 \text{ K}$

$T_{\text{sur}} = 300 \text{ K}$

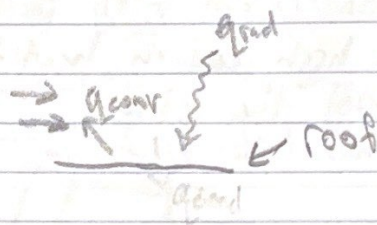
$h = 10 \text{ W/(m}^2 \cdot \text{K)}$

Find: a) Calculate the temperature of the roof

b) Would a cat be comfortable walking on the roof?

c) What is the dominant heat transfer mode

Schematic:



Properties: All properties used are listed in the known section

Assumptions: Steady state

- Bottom surface of roof is thermally insulated
- roof is a gray body

Analysis: Equation 1.10:

$$0 = k \cdot \frac{T_1 - T_2}{L} - [h(T_2 - T_\infty) + \epsilon \sigma (T_2^4 - T_{\text{sur}}^4)]$$

$$T_1 = T_2 \Rightarrow k \cdot \frac{T_1 - T_2}{L} = 0(T_1 - T_0) + \epsilon \sigma (T_2^4 - T_{\text{sur}}^4)$$

$$0.76 \cdot 750 \text{ W/m}^2 = h(T_2 - T_0) + \epsilon \sigma (T_2^4 - T_{\text{sur}}^4)$$

$$570 \text{ W/m}^2 = 10 \text{ W/m}^2 \cdot \text{K} (T_2 - 300 \text{ K}) + 0.18 \cdot 5.67 \cdot 10^{-8} \text{ W/(m}^2 \cdot \text{K}^4)$$

$$0 = T_2 \cdot 10^{-8} \text{ W/m}^2 \cdot \text{K}^4 \cdot T_2 + 3000 \text{ W/m}^2 = 1.0 (T_2^4 - 300 \text{ K}^4) T_2$$

$$570 \text{ W/m}^2 = 10 \text{ W/m}^2 \cdot \text{K} \cdot T_2 + 1.02 \cdot 10^{-8} \text{ W/m}^2 \cdot \text{K}^4 \cdot T_2^4 - 3000 \text{ W/m}^2 - 82.67$$

$$1.021 \cdot 10^{-8} \text{ W/m}^2 \cdot \text{K}^4 T_2^4 + 10 \text{ W/m}^2 \cdot \text{K} \cdot T_2 = 33082.67 \text{ W/m}^2$$

$$3652.67 \text{ W/m}^2 = 10 \text{ W/m}^2 \cdot \text{K} \cdot T_2 + 1.02 \cdot 10^{-8} \text{ W/m}^2 \cdot \text{K}^4 \cdot T_2^4$$

$$a) T_2 = 300 \text{ K}$$

$$b) T_2 = 350 \text{ K}$$

$$350 \text{ K} - 273.15 = 26.65^\circ \text{C}$$

$$b) 350 \text{ K} = 76.85^\circ \text{C} = 170.33^\circ \text{F}$$

b) A cat would not be comfortable walking on the roof because it would be 170.33°F

c) The dominant heat transfer mode is convection.